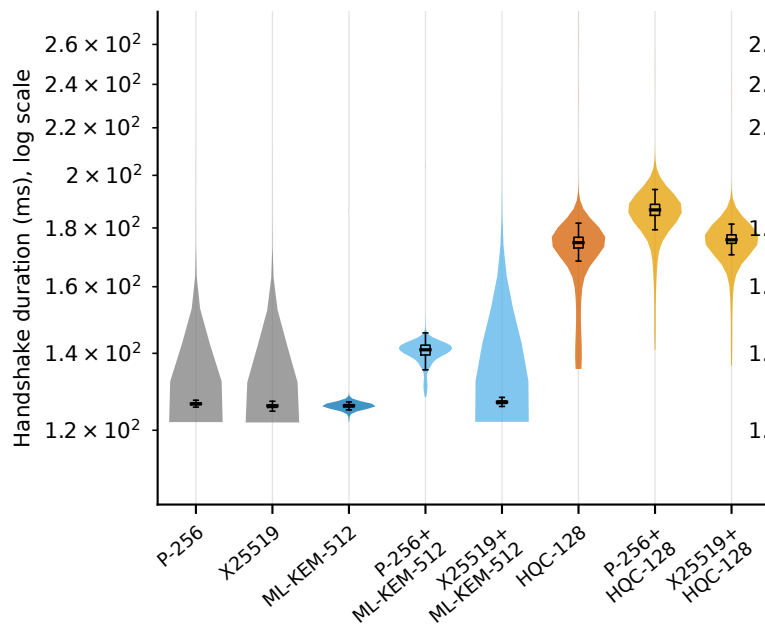
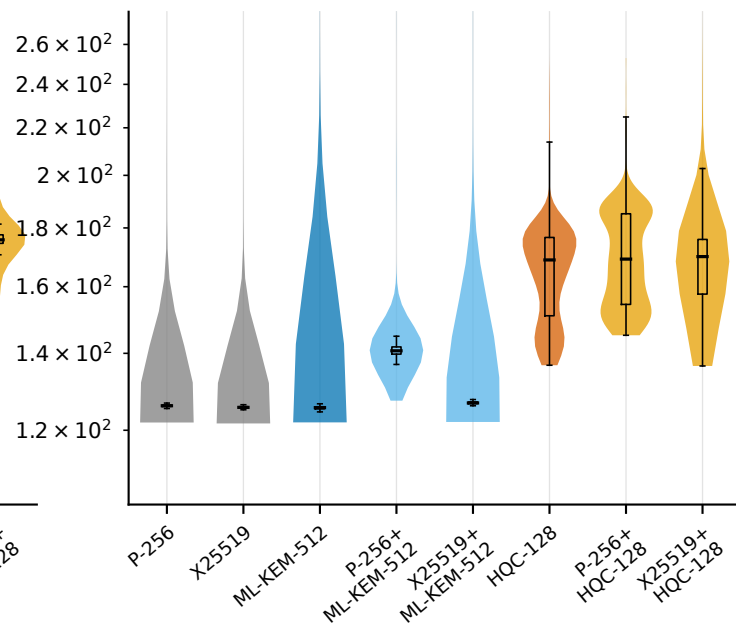


(a)

ed25519 (classical)

**(b)**

mldsa44 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)
P-256	129.1	36.6	1.15k	131.6	44.0	1.13k
X25519	129.9	39.3	1.14k	130.1	41.9	1.14k
ML-KEM-512	126.1	2.2	187.1	138.1	90.1	2.17k
P-256+ML-KEM-512	140.5	3.0	204.9	143.2	18.0	462.6
X25519+ML-KEM-512	132.6	50.9	1.15k	136.7	63.2	1.25k
HQC-128	171.5	11.8	474.5	165.7	11.3	397.0
P-256+HQC-128	186.4	11.5	481.0	170.2	9.6	253.0
X25519+HQC-128	176.4	11.0	473.5	170.2	31.5	1.19k